

Analyzing Housing Prices

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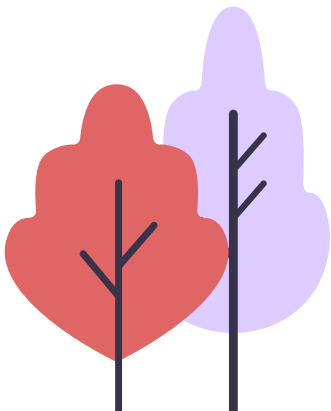


INTRODUCTION TO DATASET

The **Ames Housing dataset** contains 2,919 residential property records from Ames, Iowa from **Kaggle**. Each home is described by **79** explanatory variables across categories such as:

- Lot + Neighborhood characteristics
- House size & layout
- Exterior & material quality
- Basement, garage, and porch features
- Overall condition & quality ratings

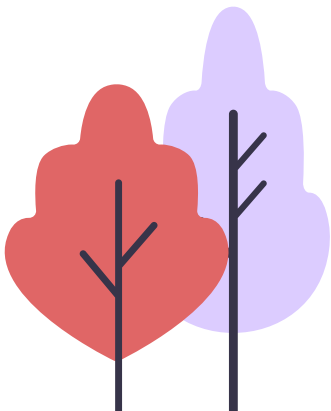
The target variable is **SalePrice**, representing the home's final selling price.



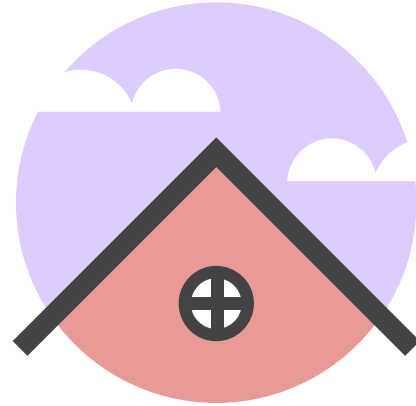


QUESTIONS

1. Can we predict housing prices based on the variables we listed earlier?
2. How did the housing crisis influence housing prices over time?
3. Does the age of home strongly correlate with lower sale prices, or do renovation offset these age effects?
4. Does the neighborhood influence the home's sale price?
5. What factors are most correlated with **SalePrice**?



PRE PROCESSING





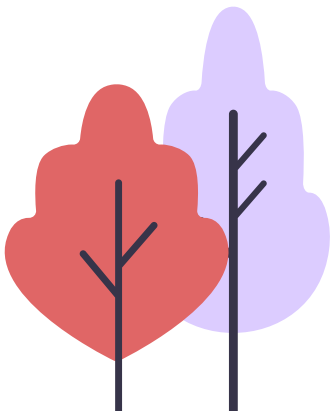
PRE-PROCESSING

- Dropped all the data which had more than 365 NA values (25% of the dataset). These ended up being

- Type of Alley Access
- Masonry Veneer Area
- Fireplace Quality
- Pool Quality
- Fence Quality
- Value of Miscellaneous Features

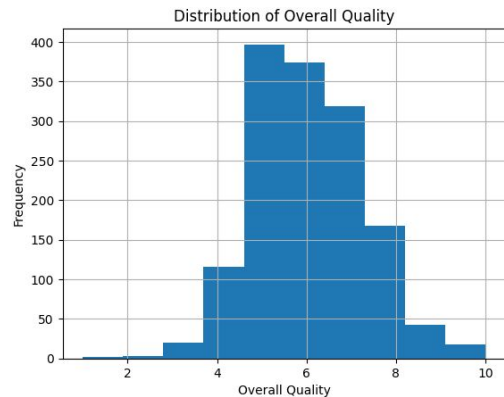
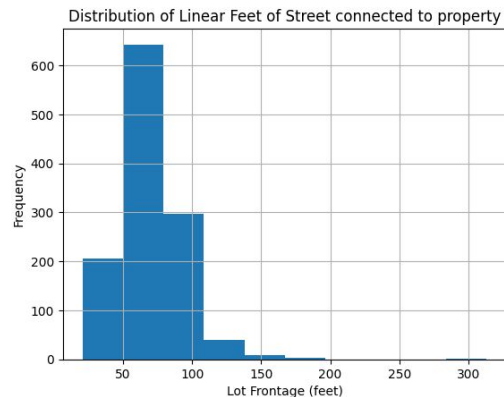
Alley	1369
MasVnrType	872
FireplaceQu	690
PoolQC	1453
Fence	1179
MiscFeature	1406

- For categorical data, we decided to use one-hot-encoding
- Categories like Zoning Classification don't have an intuitive linear scale we could use for label encoding

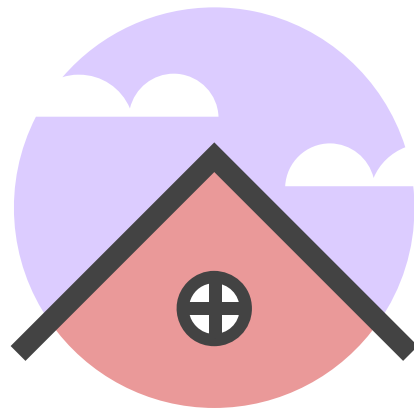


PRE-PROCESSING

- We imputing the following data with most likely values
 - Basement and Garage categorical data that were NA were imputing using the “None” category since they likely didn’t exist for these houses
 - Garage Year built that were NA were imputed as 0
- We then scaled numerical data using a Normal distribution since the training data had different orders of magnitude. Ensures linear regression weights could be easily interpreted



VISUALIZATIONS

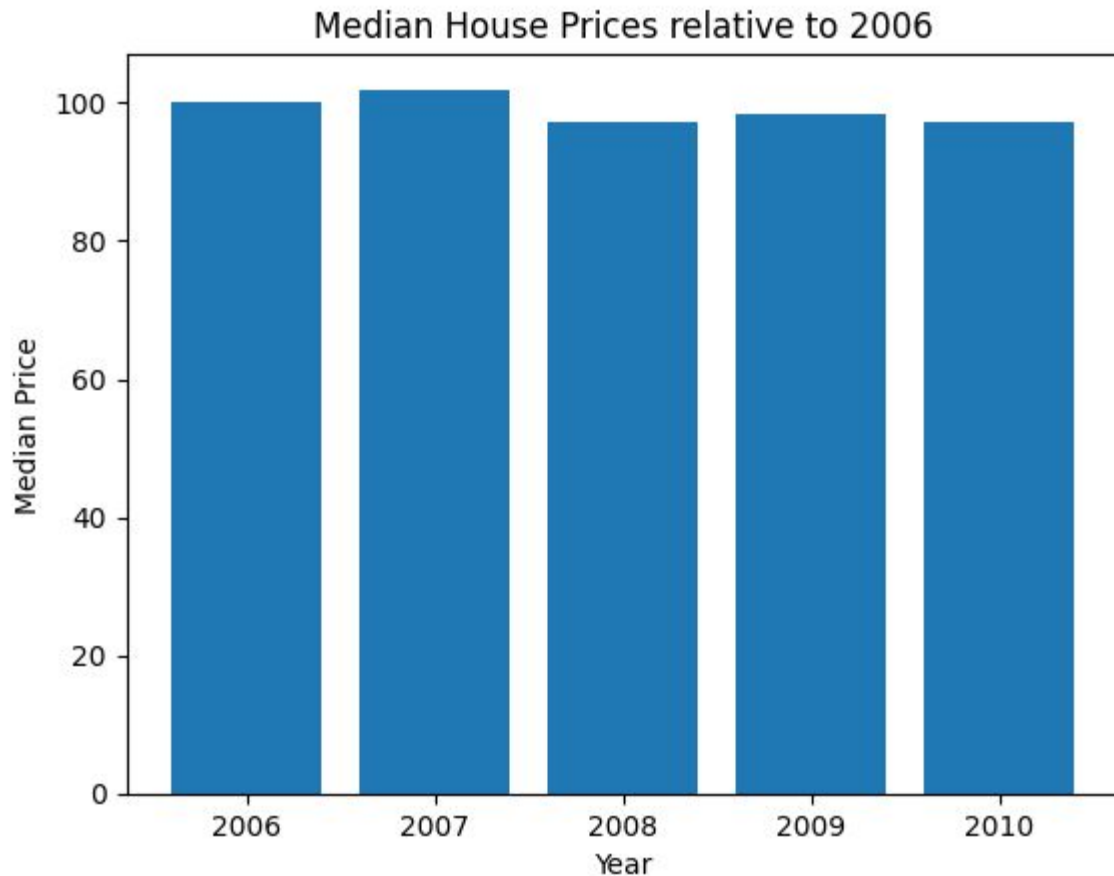


How did the housing crisis influence housing prices over time?

Short answer: Rapid drop with marginal recovery

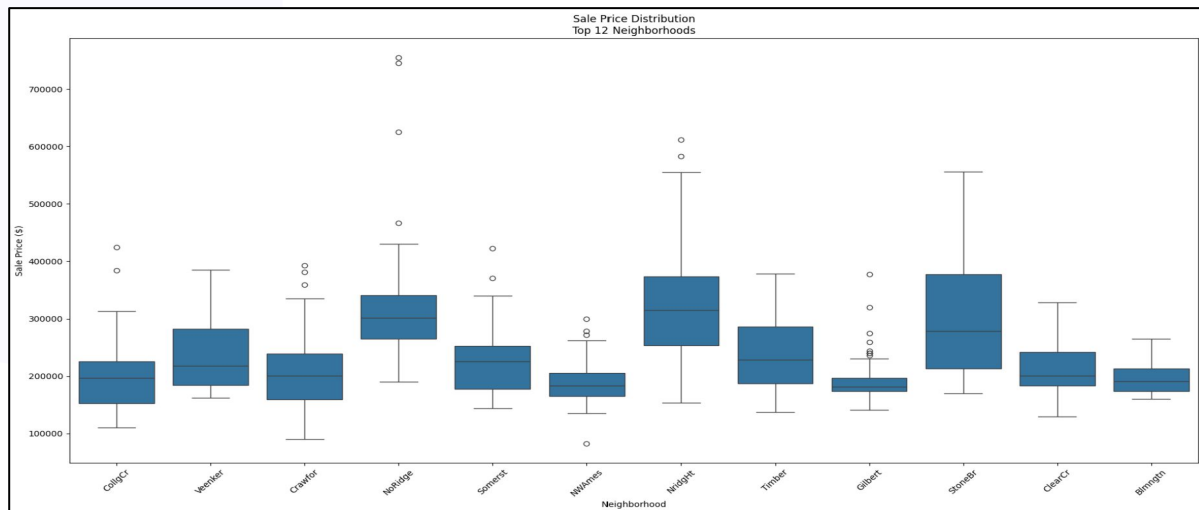
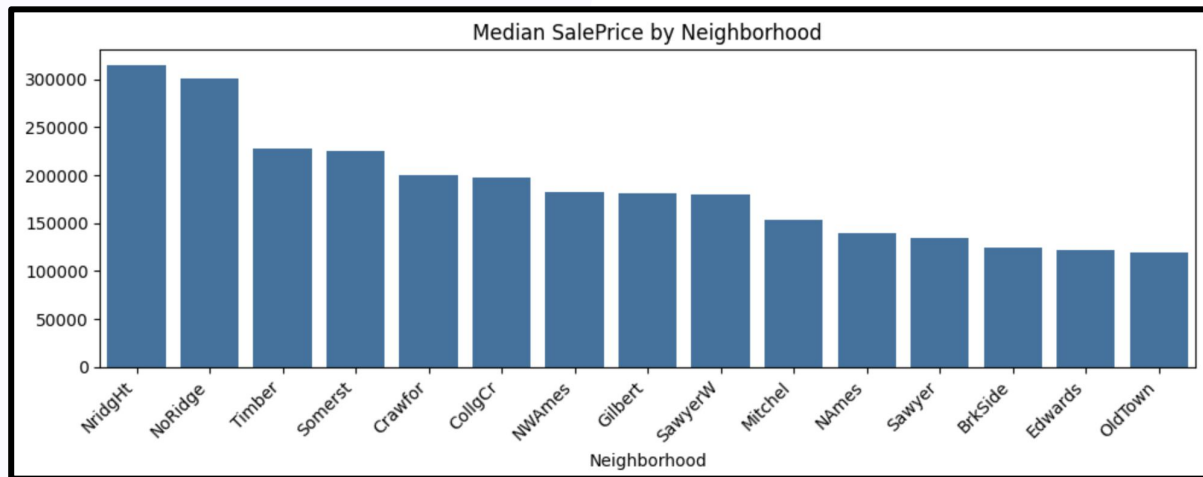
The median price of housing in 2008 was 3% below the median price in 2006

Housing Prices stagnated for the next 2 years



Does the neighborhood influence the home's sale price?

Yes, neighborhoods show large differences in both median price and overall price range.



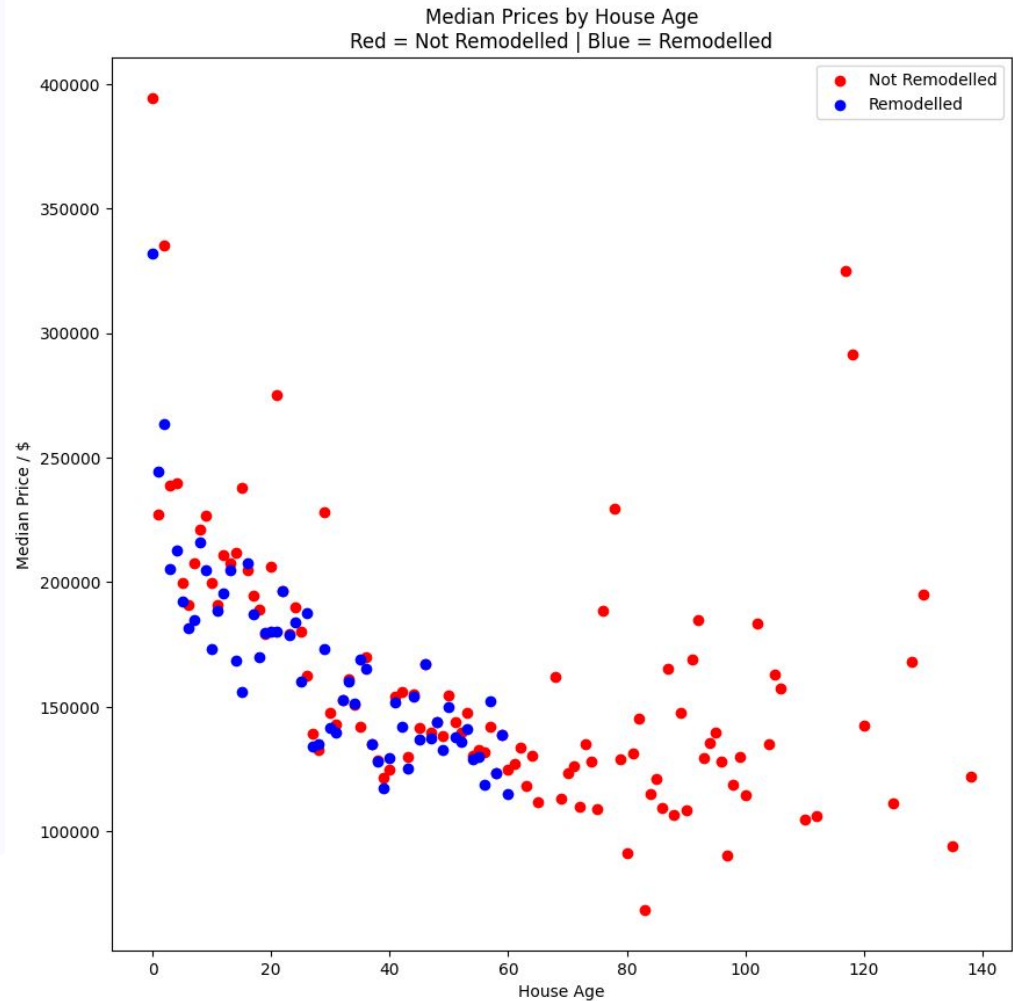
How does sales price vary due to house age (Remodeled vs. Not modeled)

Negative correlation between sales price and house age

After about 60 years, all houses get remodeled

Generally, remodeled houses had marginally lower prices than an equivalently old non remodelled house.

Houses older than 70 years gain value, resulting in V shape



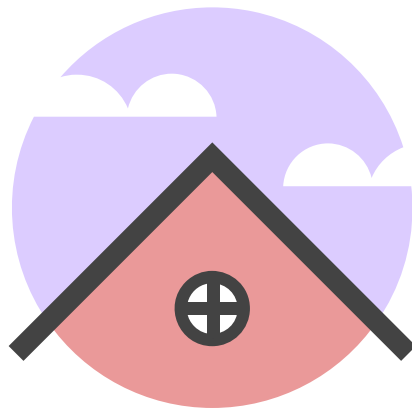
What factors are most correlated with Sale Price?

Using Regression model weights

1. House being in Low Density Residential Zoning
2. Overall Quality
3. House being in rich Neighbourhood (E.g. Crawford)
4. Houses with Central Air Conditioning
5. Amount of Above Grade Living area

MODELS

- Linear Regression
 - Ridge/Lasso Regression
- Decision Tree
- Random Forest



TESTING

We submitted our final predictions to Kaggle for official scoring and our model performs as follows:

Model	RMSLE
Ridge	0.13441
Random Forest	0.17466
Decision Tree	0.20159
Lasso	0.20355

CONCLUSIONS

- What were the biggest factors?
- What factors didn't matter so much?
- Our best RMSLE of 0.13441
- Weights best model (Ridge) were all less than 0.1
- Potential Improvements



THANK YOU FOR LISTENING!

